Report: Comparison of Language Models (RNN vs. Transformer vs. BDH) for Causal Language Modeling

1. Objective

This project implemented, trained, and compared three language model architectures (LSTM-RNN, Transformer, BDH) for next-token prediction on the WikiText-2 dataset. Models were evaluated on perplexity and time efficiency using roughly comparable parameter counts (~25-28M).

2. Setup

- **Dataset:** WikiText-2 (wikitext-2-v1 via Hugging Face `datasets`), preprocessed with basic English tokenization, a vocabulary of 28,752 tokens (min freq=3), and `<eos>` markers. Data was batched into `[sequence_length, batch_size]` format (Train: `[62421, 32]`, Val/Test: `~[13k-14k, 16]`). BPTT length was 35.

- **Hardware:** Apple M3 Pro (using MPS backend).

3. Model Architectures & Training

All models used an embedding size of 256 and a dropout rate of 0.5. Training ran for up to 15 epochs (BDH stopped early).

- **RNN (LSTM):** 3 layers, 512 hidden size. Trained with SGD (LR=15.0, ReduceLROnPlateau), gradient clipping 0.2. **~ 27.9M parameters.**

- **Transformer:** 8 layers, 4 heads, 2048 FFN hidden size. Trained with AdamW (LR=0.0002, WD=0.01, StepLR gamma=0.95), gradient clipping 0.2. **~25.3M parameters.**

- **BDH:** 4 layers, 4 heads, MLP multiplier 64 (internal dim 16384). Trained with AdamW (LR=0.0003, WD=0.01, StepLR gamma=0.95), gradient clipping 0.2. **~27.3M parameters.**

4. Evaluation Results

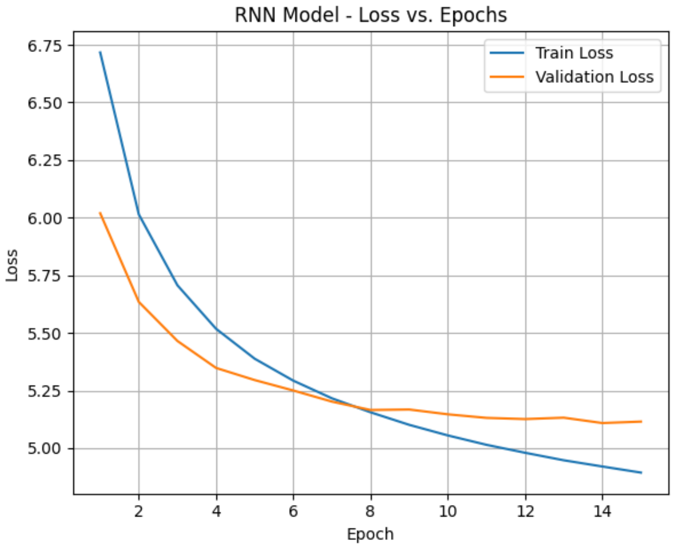
The best checkpoint based on validation perplexity was used for testing.

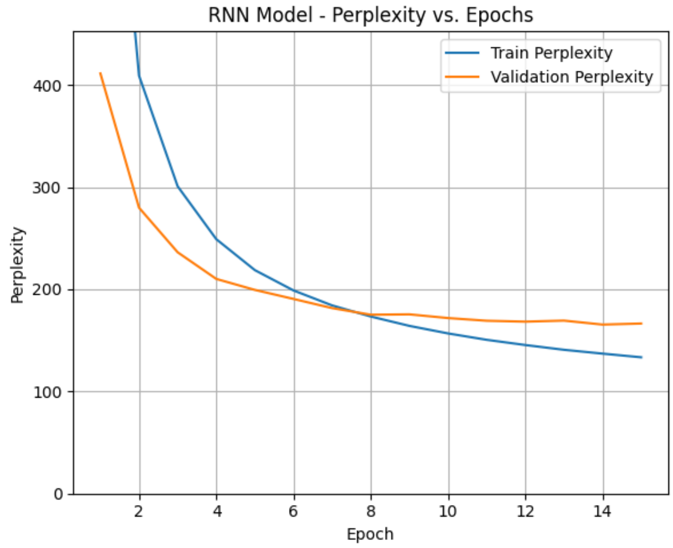
Metric	RNN (LSTM)	Transformer	BDH
Trainable Parameters	27,889,744	25,270,352	27,303,936
Best Valid Perplexity	165.48	193.12	893.03
Final Test Perplexity	**157.25**	**180.97**	**846.99**
Total Training Time (s)	1879.18	2550.82	4939.64 (4 epochs)
Inference Time (s) [10 prompts]	0.30	1.61	8.83

Loss and Perplexity Plots

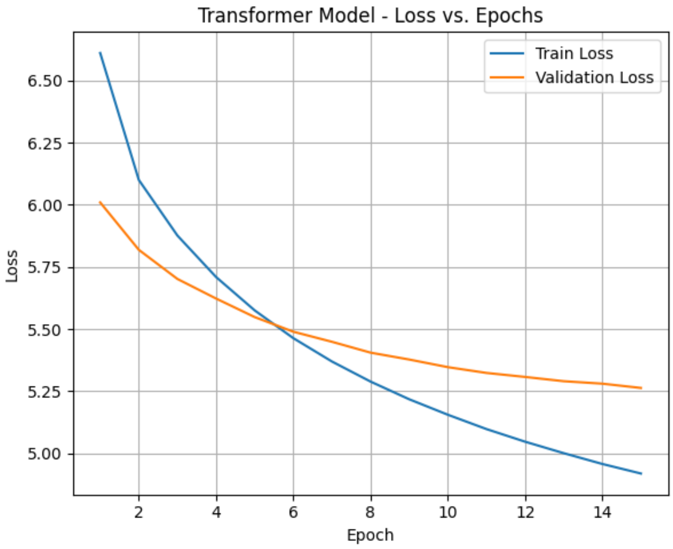
(Plots show typical learning curves for RNN/Transformer, with RNN achieving lower validation loss/perplexity faster. BDH plot shows rapid overfitting after Epoch 1.)

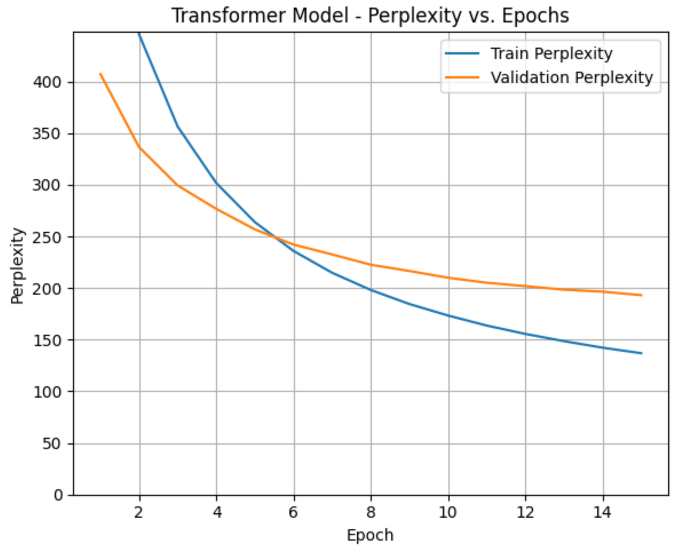
RNN Model:



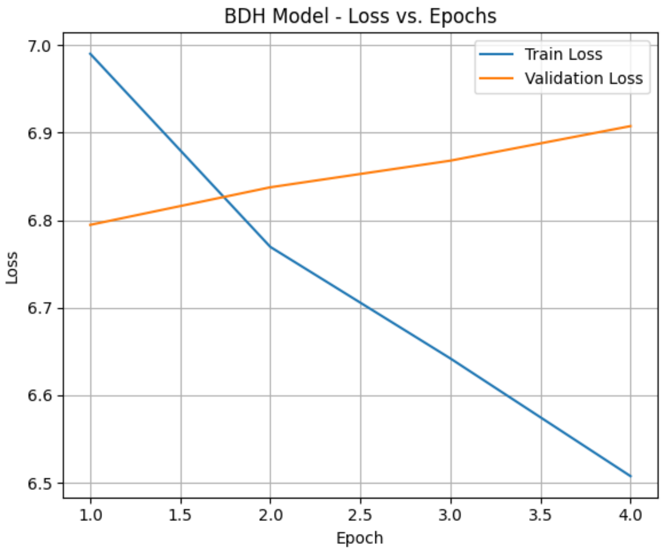


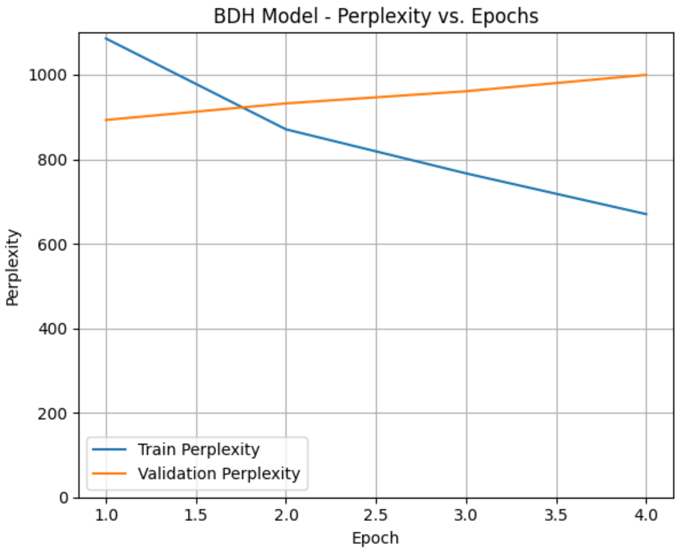
Transformer Model:





BDH Model:





5. Interpretation of Findings

- **Perplexity vs. Quality:** The RNN achieved the best test perplexity (157.25), followed by the Transformer (180.97), with BDH performing very poorly (846.99). However, qualitative analysis of generated text revealed the **Transformer produced the most coherent and least repetitive output**. The RNN frequently generated repetitive phrases or <unk> tokens, despite its lower perplexity. BDH generated primarily sequences of commas or 'the', indicating a failure to learn meaningful patterns. This highlights the limitation of perplexity as a sole metric for generative quality.
- **Time Efficiency:** The RNN was the fastest for both training and inference. The Transformer was moderately slower. BDH was extremely slow (~10x slower per epoch than RNN/Transformer) due to its large internal activation dimension (16,384), making it computationally expensive. Training times overall were long due to hardware limitations.
- **Tuning Challenges:**
    - The **RNN** required significant tuning (SGD, high LR, scheduler) to stabilize training but still yielded poor generation quality. More tuning effort was spent here compared to the Transformer.
    - The **Transformer** was prone to overfitting, requiring a high dropout rate (0.5). With limited tuning focused on the RNN, it's possible further optimization could have improved its perplexity.
    - **BDH** exhibited immediate and severe overfitting. The chosen hyperparameters were clearly unsuitable. Its high computational cost and resource constraints prevented further tuning attempts (e.g., adding stronger regularization).

6. Conclusion

In this comparison under constrained resources, the **RNN (LSTM)** offered the best perplexity score and speed but failed to generate coherent text. The **Transformer** provided the best balance, achieving reasonable perplexity with significantly better generation quality, albeit at a higher computational cost than the RNN. The **BDH** model, in this configuration, was not competitive due to immediate overfitting, extremely poor performance metrics, and prohibitive training times.